

the BeagleBoard is obviously the better choice for IoT devices. It is available with a wide range of vendors—both hardware and software are open source and sold under the name BeagleBone.

Flutter is a programmable electronics processor core for hobbyists, students and engineers. It is based on Arduino, and has a wireless transmitter with a half-mile range. It requires no router and communicates directly with other boards. It includes an asynchronous encryption of 256 bits and offers flexible usage.

LightBlue Bean Punch Through is a low power microcontroller for Bluetooth Arduino. It is wirelessly programmed, runs on a coin cell battery and is ideal for smartphone-controlled projects using Bluetooth 4.0.

Open source operating systems

AllJoyn is a free-licence OS for IoT, originally designed and developed by Qualcomm and now sponsored by the AllSeen Alliance of organisations, which includes LG, Qualcomm, Panasonic, Microsoft, Sharp, Cisco, Symantec and many others. It offers a range of services and frameworks, which enable producers to design compatible devices. It is a cross-platform API for OS X, iOS, Windows 7 and Android applications.

Contiki is an open source operating system for the Internet of Things. It merges microcontrollers to the Internet. It supports standards such as RPL, CoAP, IPv6 and 6lowpan.

Raspbian is a credit card-sized computer that is popular among developers though it was introduced as an educational device. This is not entirely open source, though much of its software and documentation is. It is the most famous Raspberry Pi OS based on Debian Linux.

Open source protocols

Advanced Message Queuing Protocol (AMQP) is a business messaging Internet protocol. It is an open standard for communicating between applications or companies. It connects systems, feeds business processes with the information they need, and transmits instructions to achieve set objectives reliably.

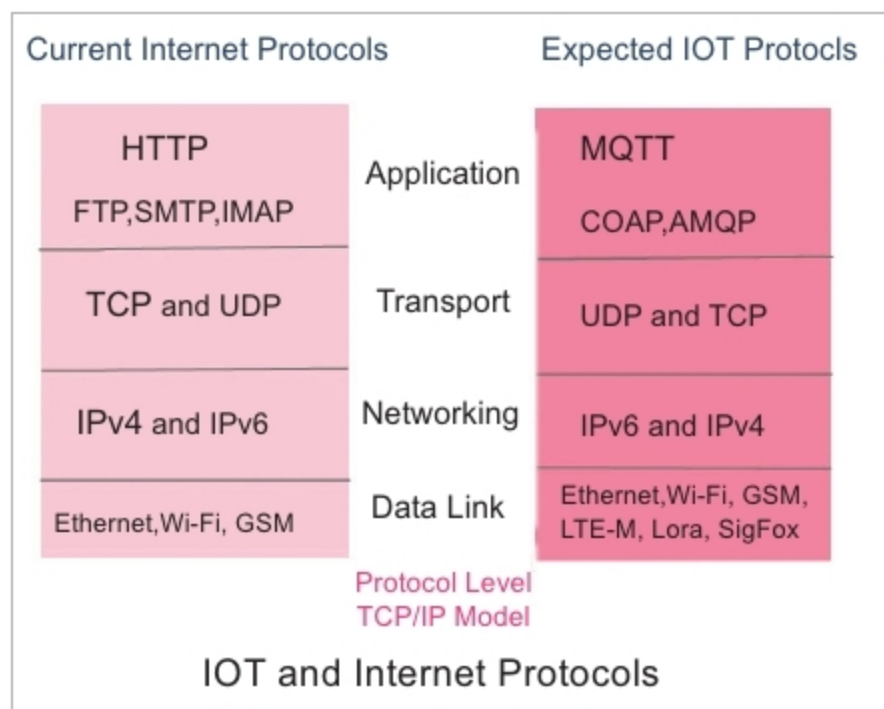


Figure 1: Internet protocol vs IoT protocol

Constrained Application Protocol (CoAP) is a “specialised Web transfer protocol to be used in the Internet of Things with restricted nodes and networks.” This protocol is intended for machine-to-machine (M2M) applications such as intelligent energy and the automation of building infrastructure.

Very Simple Control Protocol (VSCP) probably should be called a framework rather than a protocol. It is a highly scalable, very low footprint, free and open solution for the discovery and identification of devices, their configuration, autonomous device functionality, securely updating the devices — overall, a solution from the sensor to the user.

Industry consortia for open source

Open Interconnect Consortium (OIC) is a non-profit organisation founded by leading technology companies with the aim of defining the requirements for connectivity and the interoperability for the billions of devices that make up the Internet of Things.

Eclipse IoT supports open Internet of Things standards. It provides the IoT protocols open source implementations such as CoAP, oneM2M, LWM2M, MQTT, OPC-UA, and more.

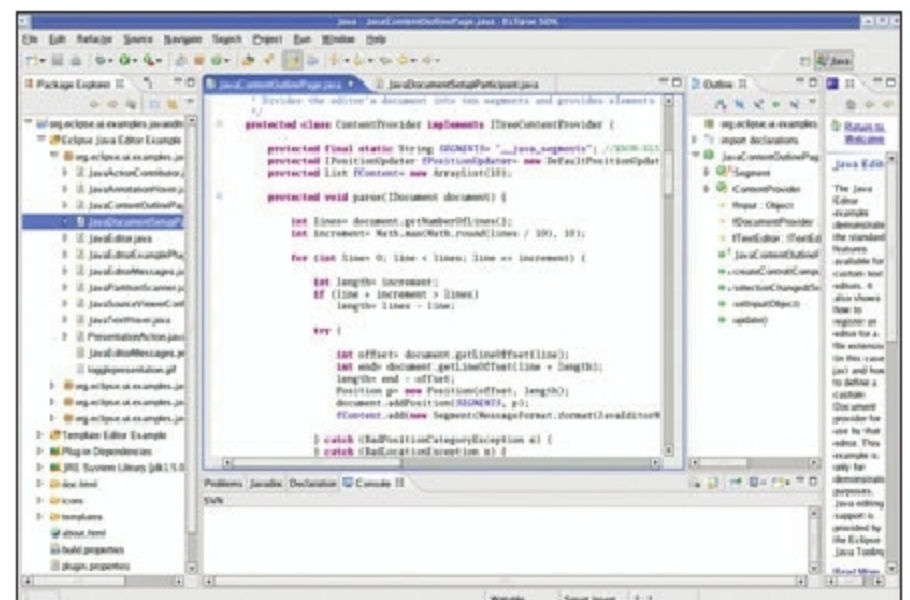


Figure 2: Eclipse IDE

Open source APIs

Zetta is an open source platform, built on Node.js to create servers running over geo-distributed computers and the cloud. Zetta combines WebSockets and REST APIs with reactive programming. This is ideal for assembling many devices in data-intensive applications in real-time.

BipIO exposes an easy-to-access HTTP API for any device. We can use its drag-and-drop simplicity to automatically connect to Web APIs with dozens of pods.

Open source middleware

IoTSyS is middleware that comes with an intelligent device communication stack. It supports several IPv6, oBIX, 6LoWPAN, Constrained Application Protocol and Efficient XML Interchange standards and protocols.

OpenIoT: Its website explains that the project is open source middleware for sensor cloud information, which saves